MIL-P-11520E
1 July 1976
SUPERSEDING
MIL-C-11520D
16 June 1961

MILITARY SPECIFICATION

PRESERVATIVE COATING, RUBBER: FOR RUBBER SURFACES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of preservative coating for the protection of natural or synthetic rubber items and is intended for spray, brush or dip application (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-X-916	-	Xylene (For Use in Synthetic Coatings).
PPP-B-636	-	Box, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-D-729	-	Drums: Metal, 55-Gallon (For Shipment of Non-Corrosive Material).
PPP-P-704	-	Pails; Shipping, Steel (1 trough 12 Gallon).

Military

MIL-P-116	-	Preservation, Methods of.

FSC-8030

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STANDARDS

Federal

Fed. Test Method Std. No. 141	-	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Federal Std. No. 356	-	Commercial Packaging of Supplies and Equipment.
Federal Std. No. 595	-	Colors.

Military

MIL STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM:

D1149	-	Test for Accelerated Ozone Cracking of Vulcanized Rubber.
D-518	-	Test for Resistance to Surface Cracking of Stretched Rubber Compounds.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Phildelphia 3, Pa.)

Rule 66 - LOS ANGELES AIR POLLUTION CONTROL

(Obtain copies from Los Angeles Air Pollution Control District, Los Angeles, California.)

3. REQUIREMENTS

3.1 Qualification. The preservation coating furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein and has been listed on or approved for listing on the applicable Qualified Products List (see 4.2).

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3.2 Materials. The solids and solvent materials required for the preservative coating shall be as selected by the contractor, within the limitation of this specification, and shall be formulated to meet the requirements specified herein (see 6.3).

3.2.1 Solvent. Solvent used to meet the requirements of this specification, shall be of the organic type and shall contain no benzene or chlorinated hydrocarbons. Where applicable, solvent shall conform to Rule 66 of The Los Angeles Air Pollution Control District.

3.3 Colors. A color shall be specified (see 6.2). Color of the coating material shall reasonably match color chip No. 27038 black, No. 24052 Marine Green or No. X24087 olive drab, conforming to Federal Std. No. 595.

3.4 Properties of preservative coating material.

3.4.1 Homogeneity. When tested as specified in 4.4.1, the preservative coating material shall show no curdling, livering, agglomeration or pigment flotation and shall show no settling or caking that cannot be brought back to a homogeneous state by manual stirring. The color shall reasonably match the color chip as specified on the contract or order.

3.4.2 Viscosity. When tested as specified in 4.4.2, the preservative coating material shall pass through the orifice of the cup in not more than 100 seconds.

3.4.3 Drying time. When tested as specified in 4.4.3, the preservative coating shall air dry to such an extent that the lower panel will not be lifted at all by the upper panel, or should there be some sticking in evidence, the coating will not pull loose from either rubber panel, when the two panels are gently pulled apart.

3.4.4 Working properties. When tested as specified in 4.4.4, and when sprayed, the applied coating shall show no more than a very slight cobwebbing and shall be opaque. When sprayed, brushed, or dipped, the applied coating shall be free of air bubbles and shall show no sagging, streaking, pinholing or other defects.

3.4.5 Low temperature flexibility. When tested as specified in 4.4.5, the preservative coating material shall be free of cracks or chips.

3.4.6 Accelerated storage. When tested as specified in 4.4.6, the preservative coating material shall be free from gelation, hard caking or other visible signs of deterioration and shall conform to 3.4.1.

3.4.7 Ozone resistance. When tested as specified in 4.4.7.3 the preservative coating and the rubber specimen underneath the coating shall be free of cracks when examined under 7 power magnification.

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3.4.8 Warehouse storage. When tested as specified in 4.4.8, the preservative coating material shall comply with the homogeneity requirements (see 3.4.1) after one-year storage.

3.4.9 Outdoor exposure. When tested as specified in 4.4.9, there shall be no cracking of the preservative coating and no cracking of the rubber specimen underneath the coating, when examined under 7 power magnification.

3.4.10 Adhesion. When tested as specified in 4.4.10, there shall be no peeling or separation of the coating from the rubber specimen.

3.4.11 Dilution stability. When tested as specified in 4.4.11, the preservative coating material shall remain stable and uniform, showing no precipitation, curdling or separation. Slight pigment settling shall be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification testing. A batch of 5-gallons of preservative coating, of each specified color (see 3.3) shall be submitted for qualification testing. The sample shall be representative of the coating proposed to be furnished to the Government. Qualification testing shall be conducted by the manufacturer, under Government surveillance (see 6.5) and shall consist of examination for the defects specified in 4.5.5.2, and testing as specified in table I.

Table I. Order of qualification testing

Test paragraph	Test name
4.4.1	Homogeneity
4.4.2	Viscosity
4.4.3	Drying time
4.4.4	Working properties
4.4.5	Low temperature flexibility
4.4.6	Accelerated storage
4.4.7	Ozone resistance
4.4.8	Warehouse storage
4.4.9	Outside exposure
4.4.10	Adhesion
4.4.11	Dilution stability

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4.2.1 Failure. Failure of a qualification sample to pass any of the examinations or tests specified herein may be cause for the Government to refuse to conduct additional examinations or testing until the faults revealed have been corrected.

4.3 Acceptance inspection.

4.3.1 Sampling for preservative coating.

4.3.1.1 Lot formation. A lot shall consist of all preservative coating material, of the same composition, manufactured at one time and in one batch, and submitted at one time for acceptance.

4.3.1.2 Sampling for acceptance testing. From each lot submitted for acceptance, one unopened container shall be selected at random.

4.3.2 Testing inspection for preservative coating.

4.3.2.1 Acceptance test. Samples selected in accordance with 4.3.1.2 shall be subjected to the test specified in table II.

Table II. Order of acceptance testing

Test paragraph	Test name
4.4.1	Homogeneity
4.4.2	Viscosity
4.4.3	Drying time
4.4.4	Working properties
4.4.5	Low temperature flexibility
4.4.6	Accelerated storage
4.4.7	Ozone resistance
4.4.10	Adhesion
4.4.11	Dilution stability

4.3.2.2 Failure. Failure of the preservative coating material to pass any specified acceptance test, may be cause for the Government to reject the lot, until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.3.3 Inspection of preservative coating. Inspection of preservative coating material, shall be in accordance with Method 1011 of Fed. Test Method Std. No. 141.

4.4 Test procedures.

4.4.1 Homogeneity. An unopened container of preservative coating material, as received and without agitation, shall be opened and examined for separation of ingredients. The preservative coating material shall then be thoroughly stirred, then examined for conformance to homogeneity and color as specified in 3.4.1.

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4.4.2 Viscosity. To determine conformance to 3.4.2, the viscosity shall be measured with a No. 4 Ford Cup when the preservative coating material is at a temperature of 77° ± 2°F.

4.4.3 Drying time. To determine conformance to 3.4.3 two panels, each 1 by 6 inch, made from test rubber (see 4.4.7.1) shall be dip coated with preservative coating using a Fisher Payne Dip Coater (or equal). The panels shall be immersed in the preservative coating and immediately withdrawn at a rate of 2 inches per minute. The panels shall be air dryed for 30 minutes at a temperature of 77° ± 2°F. The panels shall then be placed with the long edge adjacent to the edge of a table. The upper panel shall be placed at a right angle to the lower panel with at least 1 inch protruding beyond the edge of the table. A 100 gram weight shall be applied on the upper panel at the intersection for 60 seconds. At the end of the 60 second period, the weight shall be removed and the upper panel lifted gently by the protruding end.

4.4.4 Working properties.

4.4.4.1 Spray gun. To determine conformance to 3.4.4, the preservative coating material shall be applied at a spraying viscosity onto a 12 by 12 inch clean glass panel, held in a horizontal position, to a dry film thickness of 3 ± .25 mils, measured after 24 hour drying period. The application shall be by means of a spray gun (see 6.4), using an air pressure of not less than 65 pounds per square inch (psi), a fluid line pressure of not more than 5 psi, at a distance of not less than 8 inches nor more than 12 inches between the spray gun and glass panel. After air drying for 24 hours, the film shall be examined. Light from a 60 watt lamp shall be held 6 inches behind the panel to determine opaqueness.

4.4.4.2 Brushing. To determine conformance to the brushing properties of 3.4.4, the preservative coating material shall be applied to a panel of clean test rubber (see 4.4.7.1), 6 by 3 inches, with a 3-inch varnish brush. After air drying for 30 minutes, at a temperature of 77° ± 2°F., the film shall be examined.

4.4.4.3 Dipping. To determine conformance to the dipping properties of 3.4.4, the preservative coating material shall be applied to a panel of clean test rubber (see 4.4.7.1) 6 by 1 inch, by dipping (see 4.4.7.3) then examined.

4.4.5 Low temperature flexibility. Using clean test rubber (see 4.4.7.1), three specimens 6 by 1 inch shall be cut. The preservative coating material shall be applied as specified in 4.4.7.3. The specimens shall then be placed in a cold chamber and maintained at minus 20° ± 2°F. for 5 hours. At the end of the 5-hour period, while still at the specified temperature, the specimen shall be bent 180 degrees around a 1 inch mandrel. The specimen shall then be bent in the opposite direction and the bending repeated for a total of six bends in each direction within a period of 20 seconds. The film of at least two of the three specimens shall pass the examination for conformance to 3.4.5.

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4.4.6 Accelerated storage. To determine conformance to 3.4.6, a 1-pint, extra tight friction lid, metal container (terneplate excluded) filled to 75 percent of capacity with the preservative coating material shall be sealed and stored in a temperature-controlled oven for 7 days at 120° ± 5°F.

4.4.7 Ozone resistance.

4.4.7.1 Test rubber. The preservative coating material shall be applied on test rubber conforming to the following composition:

Materials	Parts by weight
SBR-1500	100
Zinc oxide	3
Stearic acid	2
Phenyl-b-napthylamine	1
Polymerized trimethyl dihydro quinoline	1
HAF type carbon black	50
N-cyclohexyl-2-benzothiazyl sulfonamide	1
Sulfur	1.75

The test rubber shall be cured into sheets 0.075 inch in thickness, using a minimum pressure of 500 psi at a temperature of 307°F. for 30 minutes.

4.4.7.2 Apparatus. The apparatus shall be similar to the apparatus described in ASTM-D1149.

4.4.7.3 Procedure. Using clean test rubber (see 4.4.7.1), three 6 by 1 inch specimen shall be cut. The preservative coating material at a temperature of 77° ± 2°F., shall be applied by dipping once. Dipping shall be accomplished by immersion of test specimen in coating material, withdrawing partially, then completely immersing specimen again and withdraw rapidly. Let dry for a period of 24 hours. The coated specimen shall then be elongated 25 percent and conditioned for 45 minutes at room temperature. The elongated specimens shall then be placed in an ozone chamber and exposed to a mixture of ozone and air in the proportion of 50 ± 5 parts of ozone per 100 million parts of air by volume, for 7 days at a temperature of 100° ± 2°F. The specimens shall then be removed and examined for conformance to 3.4.7.

4.4.8 Warehouse storage. A 1-pint sample of preservative coating material shall be stored under normal warehouse conditions at the qualifying laboratory for 1 year, and examined for conformance to 3.4.8.

4.4.9 Outdoor exposure. To determine conformance to 3.4.9, preservative coating shall be subjected to an outdoor exposure test, for a period of one year, at the qualifying laboratory, in accordance with method B of ASTM-D518.

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4.4.10 Adhesion. An apparatus shall be provided to rapidly elongate a specimen from 0 to 100 percent. Specimens that have passed the ozone test (see 4.4.7) shall be elongated from 0 to 100 percent, 100 times in approximately 3 minutes, at room temperature and examined for conformance to 3.4.10.

4.4.11 Dilution stability. To determine conformance to 3.4.11, reduce the preservative coating material with one part by volume of xylene thinner, conforming to TT-X-916. The material shall then be tested in accordance with Method 4203 of Fed. Test Method Std. No. 141.

4.5 Inspection for preparation for delivery.

4.5.1 Materials and processes. The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery to determine conformance to requirements of section 5 and specifications referenced therein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided that corrective action has been taken.

4.5.2 Lot formation. A lot shall consist of all packages and packs prepared for shipment in accordance with one level (see 5.1 and 5.2) from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.3 Sampling for acceptance examination. Samples for acceptance examination shall be selected in accordance with MIL-STD-105.

4.5.4 Sampling for acceptance testing. One package, and one complete pack, when using 1-gallon containers or one pack when using 5-gallon pails or 55-gallon drums, shall be selected from each lot for acceptance testing.

4.5.5 Examination inspection for packaging and packing.

4.5.5.1 Acceptable quality level. Each sample selected in accordance with 4.5.4.3 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.5.2 Classification of defects. Examination shall be conducted as specified in table III. Any sample containing one or more defects shall be rejected, and if the number of defects in any one sample exceeds the acceptance for that sample, the lot represented by that sample shall be rejected.

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Table III. Classification of defects in packaging and packing in accordance with Standard MIL-STD-105

Categoreis	Defects
Major:	
101	Improper degree of preservation and packing (see 6.2).
102	Improper container size (see 5.1.1.1).
103	Containers are not clean (see 5.1.1.1).
104	Overfill or underfill (see 5.1.1.1).
105	Leakage from container in any position (see 5.1.1.1).
106	Illegible marking (see 5.3).
107	Incorrect marking (see 5.3).
Minor:	
201	Improper closure or sealing of container (see 5.1.1.1).

4.5.6 Testing inspection for the package and pack

4.5.6.1 Acceptance testing. When level A or B is specified (see 6.2), the samples selected in accordance with 4.5.4.4 shall be subjected to the preservation examination and packing tests specified in MIL-P-116 or MIL-STD-1186, as applicable (see 5.1.1.1, 5.2.1, 5.2.2).

4.5.6.2 Failure. Failure of the package, or pack to pass any specified test, may be cause for the Government to refuse to accept the lot until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

5. PACKAGING FOR DELIVERY

5.1 Preservation. Preservation shall be level A, B or commercial as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Unit package. When specified (see 6.2), the preservation coating material shall be furnished in one-gallon containers, conforming to type V, class 2 of PPP-C-96, or 5-gallon containers, conforming to type II, class 3 of PPP-P-704, or 55-gallon containers equipped with a double blade agitator, conforming to type IV of PPP-D-729. When tested as specified in 4.5.6.1, the unit package shall meet the examination requirements specified for method III of MIL-P-116.

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5.1.2 Commercial packaging. When specified (see 6.2) the preservative coating material shall be furnished in 1-gallon, 5-gallon or 55-gallon containers and unit packing shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Level A. The 1-gallon container shall be packed four containers to a box, conforming to type 1, class 2, style RSC of PPP-B-636. Gross weight shall not exceed the weight limitation specified in PPP-B-636. The box shall be closed and secured as specified in the appendix thereto. The 5-gallon and 55-gallon containers shall be used as the ultimate shipping containers. When tested as specified in 4.5.6.1, the shipping containers shall meet the requirements of the interior packing test of MIL-STD-1186.

5.2.2 Level B. The 1-gallon container shall be packed four containers to an exterior container conforming to those listed in paragraph 5.2.1 and shall be class 1 instead of class 2. Gross weight and closure shall be as specified in paragraph 5.2.1. The 5-gallon and 55-gallon containers shall be used as the ultimate shipping containers. Testing shall be as specified in paragraph 5.2.1.

5.2.3 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129. For levels A and B and in accordance with FED-STD-356 for commercial.

6. NOTES

6.1 Intended use. The preservative coating covered by this specification is intended for use on such rubber items as tires, track blocks, bogie wheels, gaskets, hose, and other such rubber materials exposed to the weather in outdoor storage. The material can be used where solvent vapors are not a fire hazard. The material is suitable where storage temperatures may be below 32°F. Hazardous materials will be handled in accordance with (OSHA) Occupational Safety and Health Act, Standards.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color shall be specified (see 3.3).
(c) Degree of preservation (see 5.1) and packing (see 5.2).
(d) Size of container required (see 5.1.1.1).

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6.3 Materials. The materials may consist of elastomeric compounds, fillers and color pigments in organic solvents. When necessary, antiozonants may be added to meet ozone resistance requirements.

6.4 Spray equipment. Spray equipment (see 4.4.4.1) used for this test may be a Graco Paint Master, equipped with a De-Vilbiss MBC spray gun, 765 spray head and FX, FF or G tip, or equal.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the product covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from Army Tank-Automotive Command, ATTN: AMSTA GSES, Warren, Michigan 48090.

6.6 Thickness of preservative coating material. The thickness of coating material specified in 4.4.4.1 is for controlled laboratory test purposes only and should not be construed as the thickness requirements for field application, which is not a part of this specification.

Custodians:
Army - AT
Navy - SH

Review:
Army - EL, GL, MR, SM
Navy - AS

User:
Navy - MC, SA, OS

Preparing activity:
Army - AT

Project No. 8030-0429

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved use of the specification in procurement of products for ultimate use by the Department of Defense. This is provided for obtaining information on the use of this specification which will insure that suitable pro can be procured with a minimum amount of delay and at the least cost. Comments and the return of this will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Co and suggestions submitted on this form do not constitute or imply authorization to waive any portion o referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-P-11520E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROC MENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional attach to form and place both in an envelope addressed to preparing activity)*

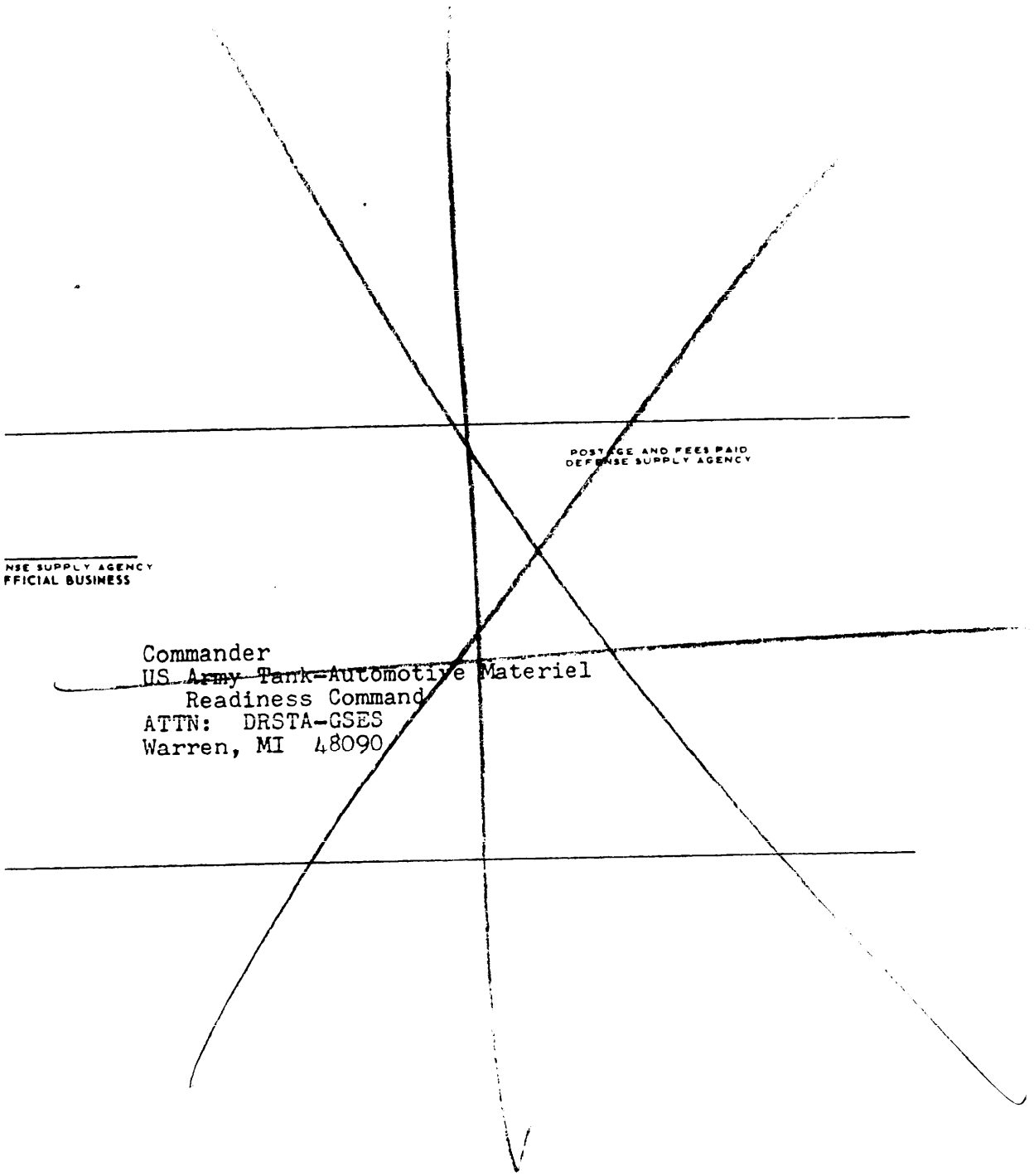
SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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